Purna Dutta

Email | Homepage | GitHub

PhD researcher at UMass Amherst with a strong foundation in theoretical computer science and intermediate experience in machine learning and large-scale experimental evaluation.
Academic email: purnadutta@umass.edu

Education

University of Massachusetts Amherst — **Amherst, MA, USA**
PhD, Theoretical Computer Science — *2024 – 2029 (expected)*
Advisors: Prof. Hung Le and Prof. Andrew McGregor.
GPA: 3.90/4.00
Research in graph algorithms, dynamic streaming, and nearest-neighbor search.

Indian Institute of Science Education and Research (IISER) — **Berhampur, India**
BS-MS, Mathematics and Computer Science — *2019 – 2024*
CPI: 9.30/10.00
MS thesis: *Expander Decomposition and Applications*.
Advisor: Prof. Saket Saurabh.
Thesis hosted by the Institute of Mathematical Sciences (IMSc), Chennai, May 2023 – April 2024.

- Reviewed efficient ways to split graphs into well-connected parts and their uses.
- Took graduate courses in algorithms, discrete mathematics, logic, and optimization at IMSc.
- Technical skills: algorithm design and analysis, expander decomposition, graph algorithms, cut sparsification, approximation algorithms, combinatorial optimization.

Publications

On Worst-Case Guarantees for Graph-Based Nearest-Neighbor Search. *NeurIPS 2026*.
Majid Daliri, Mohammadreza Daneshvaramoli, *Purna Dutta*, Edo Liberty, Cameron N. Musco, Christopher Musco, Torsten Suel.

Vertex-Fault-Tolerant Spanners and Sparsifiers in Dynamic Streams. *Under submission*.
Purna Dutta, Hung Le, Andrew McGregor.

Honors & Awards

2025: **James Kurose Scholarship in Computer Science**, UMass Amherst; part of the department's donor-funded scholarship program.

2024: **Joint Entrance Screening Test (JEST)**: All India rank 55 in Part A; one of 16 applicants across India to qualify Part B.

2022: **DAAD-WISE Scholarship**: Three-month research fellowship from the German Academic Exchange Service. Awarded to fewer than 12% of applicants.

2021: **IASc-INSA-NASI Summer Research Fellowship**: two-month research fellowship from the three Indian science academies; awarded to fewer than 10% of applicants.

2019 – 2024: **INSPIRE-SHE (Scholarship for Higher Education)**: Five-year Government of India scholarship for top-1% performance nationally in school-leaving exams.

2019 – 2023: **Outstanding undergraduate coursework**: Received O (outstanding) grades in 10 undergraduate courses.

Research Experience

University of Massachusetts Amherst

Navigable Graphs for Approximate Nearest-Neighbor Search *2025 – 2026*

Paper accepted at NeurIPS 2026

- Proved limits on the speed of similarity search using graph-based methods.
- Developed models to analyze the running time of practical search methods.
- Built an open-source framework to test changes to search graphs. **[Code]**
- Compared search accuracy and the number of comparisons needed on several benchmark datasets.
- Technical skills: algorithm design and analysis, computational complexity, query-time lower bounds, high-dimensional nearest-neighbor search, Steiner-point augmentation, experimental design, benchmarking.

University of Massachusetts Amherst

Dynamic Streaming Algorithms for Fault-Tolerant Graph Sparsification *2024 – 2026*

- Designed algorithms to summarize massive, changing networks while allowing for node failures.
- Analyzed tradeoffs between memory use, speed, accuracy, and tolerance of node failures.
- Technical skills: algorithm design and analysis, randomized and streaming algorithms, fault-tolerance, graph sparsification, probabilistic analysis, approximation algorithms, computational complexity, streaming-space complexity.

Technische Universität Berlin **Berlin, Germany**

DAAD-WISE Research Fellow: Machine Learning and Geospatial Data Analysis *May – Jul 2022*

Supervisor: Robert Spang, under Prof. Sebastian Möller.

- Cleaned over a month of location data, fixing errors in about 30% of the records. Errors included mismatched locations and times, overlapping timestamps, and impossible travel segments.
- Built a pipeline to clean and check the data. Helped design experiments and check results for two peer-reviewed publications [1] [2].
- Used self-recorded heartbeat data collected continuously over one week, at rest and in motion. Trained a model to distinguish these states with 85% accuracy.
- Technical skills: machine learning, data validation, feature engineering, supervised learning, neural networks, model evaluation, experimental design.

Indian Statistical Institute (ISI) **Delhi, India**

Research Intern: Statistical Analysis *May – Sep 2021*

Supervisor: Prof. S. K. Neyogi.

- Automated the collection and cleaning of live data.
- Explored patterns in the data, built statistical models, and used tests to assess the findings.
- Technical skills: statistical inference, exploratory data analysis, hypothesis testing, statistical modeling, data preprocessing.

National Institute of Science Education and Research (NISER) **Bhubaneswar, India**

IASc Summer Research Fellow: Transcendental Numbers *May – Jun 2021*

Supervisor: Prof. Brundaban Sahu.

- Studied constructions of transcendental numbers and proofs of transcendence.
- Technical skills: number theory, proof analysis, mathematical presentation.

Selected Projects

University of Massachusetts Amherst

Quantum Network Entanglement Routing *2026*

Graduate course project.

- Built a simulator for simultaneous connections between multiple pairs of users in a quantum network, extending earlier single-pair models. **[Code]**
- Compared two routing methods on three standard network layouts, modeling limited resources, unreliable links, and declining connection quality.
- Technical skills: combinatorial optimization, shortest-path routing, multi-commodity flow, stochastic simulation, quantum-network modeling, entanglement generation and swapping, memory and fidelity constraints.

Selected Projects (contd.)

University of Massachusetts Amherst

Adaptive Poker Agent

Graduate course team project.

- Co-designed a poker bot's betting rules to adapt to opponents; helped integrate and debug its components.
- In reported example matches, the team bot won all 2,000 chips against each of three comparison agents.
- Technical skills: decision-making under uncertainty, Monte Carlo estimation, adaptive heuristics, empirical evaluation, debugging, integration, optimization.

Leadership & Service

Conference reviewing: COCOON 2026; FOCS 2026.

University of Massachusetts Amherst

Student Volunteer, PhD Applicant Support Program *2024 – 2025*

Mentored four PhD applicants and reviewed their application materials through a UMass program supporting underrepresented applicants. [Program]

Mathematics Club, IISER Berhampur

Core Committee Member, Project Spiral *2020 – 2023*

- Co-organized an international seminar series with 10+ faculty speakers and 100+ attendees.
- Co-organized a π-Day lecture by Prof. Ram Murty of Queen's University.
- Co-organized student seminars on uses of mathematics in other fields.
- Co-organized an institute-wide mathematics quiz.
- Explained Goldbach's Conjecture to school students at an outreach event.
- Managed event budgets, logistics, and resources. [Club]

Teaching

University of Massachusetts Amherst

Teaching Assistant: CS 611, Design and Analysis of Algorithms *Fall 2025*

Held office hours, explained advanced algorithms, and graded exams. [Course]

University of Massachusetts Amherst

Teaching Assistant: CS 311, Algorithms *Spring 2025*

Led problem-solving sessions, held office hours, and graded assignments and exams. [Course]

Relevant Coursework

Computer science and theory: Algorithm Design and Analysis; Theory of Computation; Discrete Mathematics; Logic; Randomized Algorithms; Parameterized Algorithms; Linear Programming and Combinatorial Optimization; Programming and Data Structures; Error Analysis.
Artificial intelligence and data science: Artificial Intelligence; Reinforcement Learning; Introduction to Data Science.
Mathematics: Combinatorics and Graph Theory; Optimization Techniques; Calculus; Group Theory; Probability; Linear Algebra; Numerical Analysis; Elementary Number Theory.
Quantum: Quantum Communication: quantum information, mixed states, entanglement and entanglement generation, quantum key distribution, long-distance quantum communication, quantum networks, noise, and basic quantum error correction.

Workshops & Schools Attended

2026: **Computational Geometry Week**, Rutgers University, New Brunswick, NJ.

2025: **DIMACS Workshop on Hardness of Approximation in P**, Rutgers University, NJ. Talks and tutorials on the limits of efficient algorithms.

2025: **Women in Theory**, Simons Institute, Berkeley, CA. Three-day series of talks on theoretical computer science.

2024: **Annual Fall Workshop on Computational Geometry**, Tufts University, Medford, MA. Two days of talks on selected research papers.

2023: **Advanced Instructional School on Algorithmic Graph Theory**, Indian Institute of Technology, Indore. Twenty-one days of lectures on current graph theory research.

2019: **Vijyoshi Camp for INSPIRE-SHE Scholars**, IISER Kolkata. Three-day workshop on applications of STEM.

Other Academic Activities

- Helped analyze Pan-STARRS telescope images through the International Astronomical Search Collaboration.
- Completed three Coursera courses: *Programming for Everybody (Getting Started with Python)*, *Python Data Structures*, and *Using Python to Access Web Data*.
- Attended the *Statistics and Big Data* workshop led by Dr. Rajeeva Karandikar and organized by IIT Madras for Research Scholars Day.
- Attended the RAD@home radio-astronomy workshop at IISER Berhampur in 2019.

Extracurricular Activities

- Trained in Kathak, an Indian classical dance form, for seven years.
- Published articles in newspapers, including *The Telegraph*, and in a college magazine.
- Published a poem in *What Tomorrow Brings* (The Write Order).
- Participated in literary writing competitions at institutes across India.
- Emceed college events.

References

Available upon request.